COA

26/3/26

Assignment :- 2

* Section-A

Ans-1 CPU registers are small, high-speed storage locations inside the processor

Importance :-
- Store intermediate data & instruction
- Faster than cache & memory
- Improve execution speed of Program

Ans-2 Program Counter (PC) is a register that holds the address of next instruction

Function:

- Points to next instruction to be executed
- Automatically updates after each execution
- Controls instruction sequence.

Ans-3: Instruction format defines the structure of an instruction in binary form.

Key fields:

- Opcode (operation)
- Operands
- Addressing Mode

Ans-4: Addressing Mode specify how to locate operands.

Purpose:
- Provide flexibility in accessing data
- Reduce instruction count
- Improve program efficiency.

Ans-5: ISA is the interface between hardware & software defining supported instructions.

Includes:-
- Instruction formats
- Data types
- Registers & addressing modes.

Ans-6 Types of instructions in a processor are:

- Arithmetic (ADD, SUB)
- Logical (AND, OR)
- Data Transfer (LOAD, STORE)
- Control (JUMP, BRANCH)

Ans-7 Explanation:

- Opcode is interpreted
- Required operation is identified
- Control signals are generated
- Operands are prepared for execution

Ans-8 Basic instruction cycle are:

- Fetch instruction from memory
- Decode instruction
- Execute operation
- Store result.

Ans-9 characteristics of RISC processors are:

- Simple & fixed-length instructions
- few addressing Modes
- faster execution (Pipeline-friendly)
- load/store architecture

Ans to Characteristics of CISC processors are:

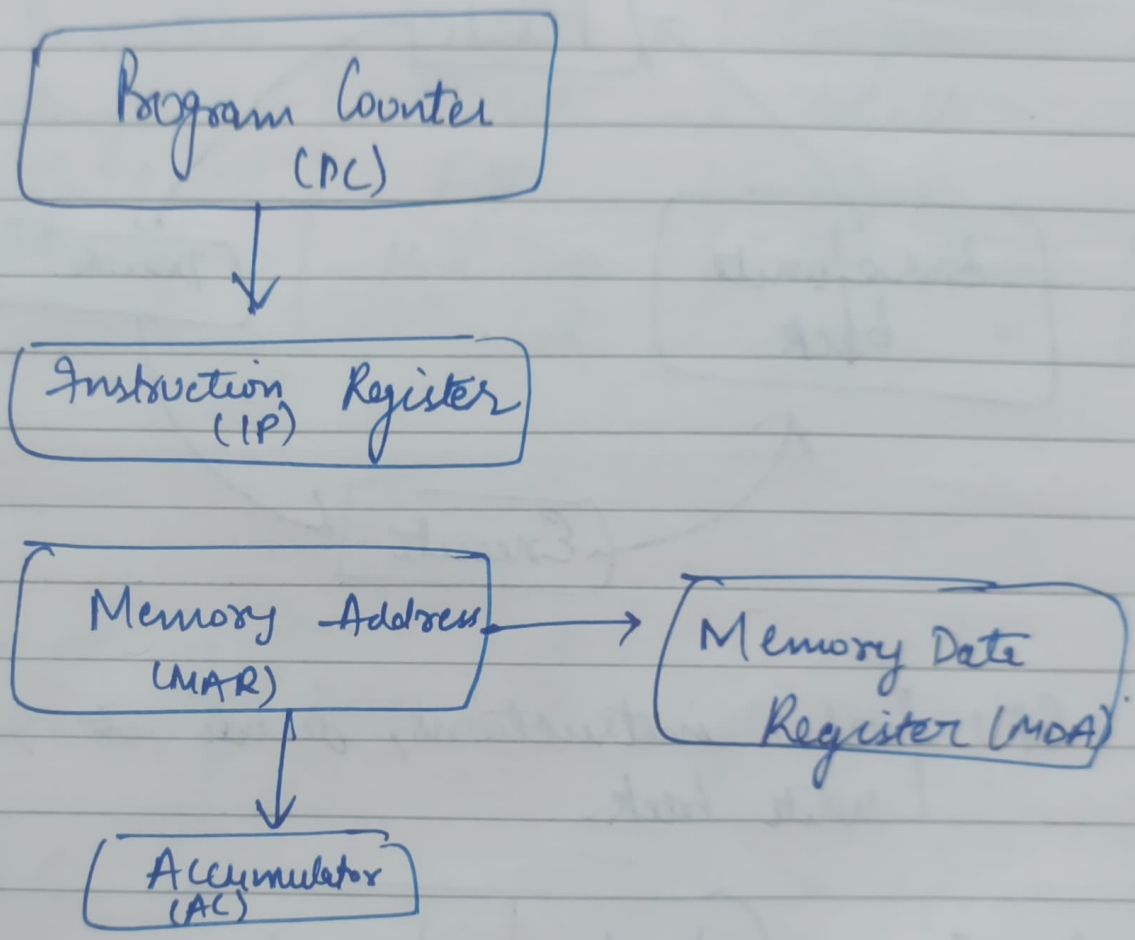
- Complex and variable length instructions
- Many addressing Modes
- fewer instructions per program.
- Slower but powerful instructions

Ans-11

RISC-V:	X86:
↳ Open-source ISA ↳ Simple, fixed-length instructions ↳ Easier to design & extend	↳ proprietary (intel/AMD) ↳ Complex, variable-length instructions

* Section-B

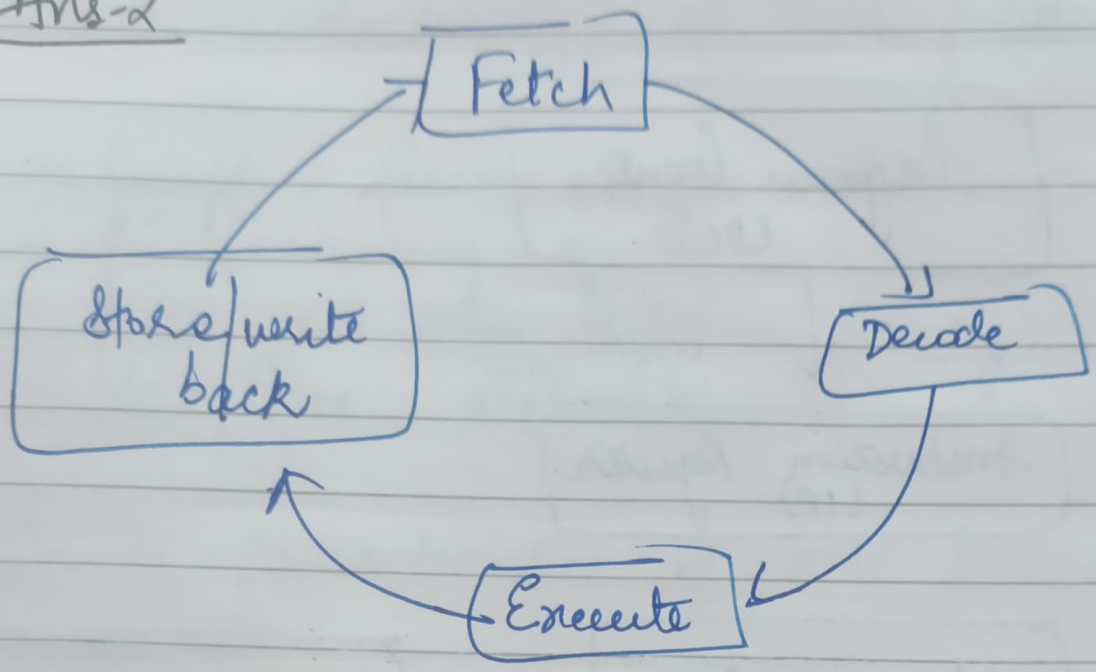
Ans-1



Explanation:

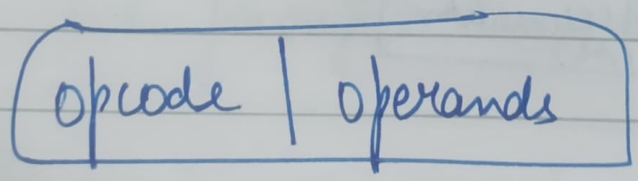
- PC holds next instruction address; IR stores current instruction
- MAR & MDR handle memory access; Account stores intermediate results.

Ans-2



CPU fetches instructions, decode it, execute it & write back.

Ans-3



Explanation:

- opcode specifies the operations
- operands specify data or memory location

Ans-4

Feature	RISC	CISC
Instruction	Simple	Complex
length	Fixed	Variable
Speed	Fast	Slower
Memory	More Usage	Less usage.

Explanation:

- RISC focuses on simplicity & speed
- CISC focuses on powerful and complexity

* Section-C

Ans-1 Impact on Performance are:-

→ Memory access is slower than register access
→ Increase CPI due to memory latency
→ Overall execution time increases → Performance ↓

Ans-2 RISC architecture should be choosen because:

→ Simple and fixed-length instructions
→ Easier to pipeline efficiently.
→ Requires simpler hardware design.

Ans-3 CPU executes the instruction as:

Step 1: Fetch
- Instruction is fetched from memory into IR

Step 2: Decode
- opcode is decoded & registers are identified.

Step 3: Execute

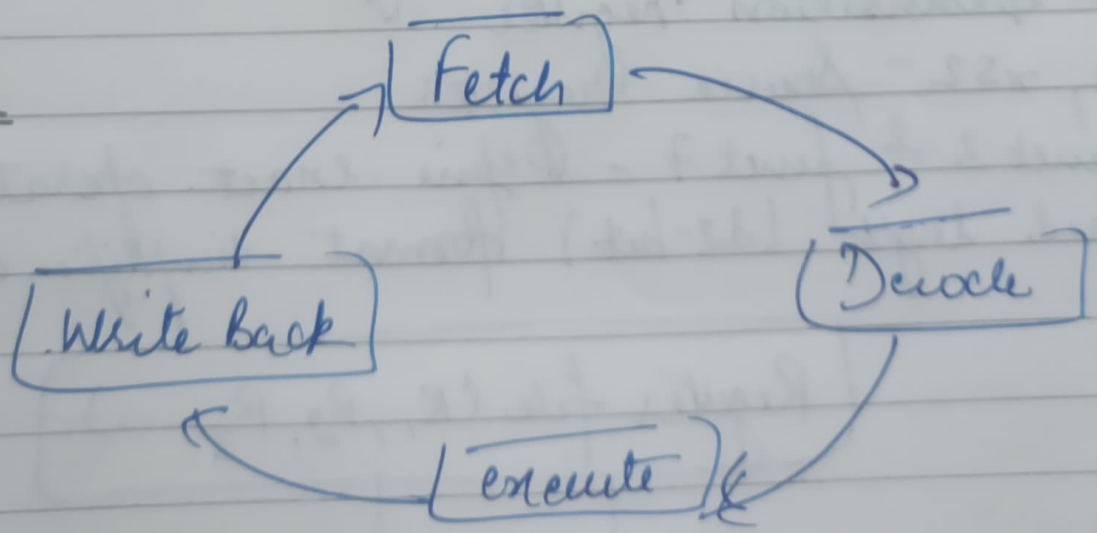
- Data is read from registers
- ALU performs operation
- Result is stored back in register.

Ans-4 X86 is widely used despite Complexity because:

→ Backward compatibility with Older Software.
→ Large ecosystem (OS, applications, industry support)
→ Modern CPUs internally optimized
→ Despite complexity, compatibility & ecosystem keep X86 dominant.

* Section-D

Ans-1



Explanation of fetch Stage:-

- PC holds address of next instructions
- Address sent to memory via MAR
- Instruction fetched into MDR and loaded IR.
- PC is incremented to next instruction

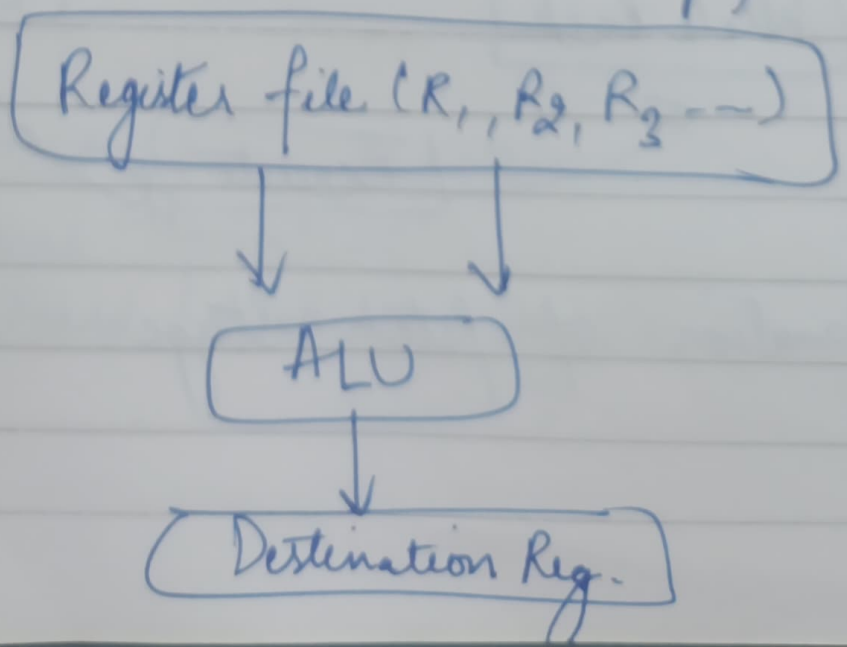
Ans-2

funct 7	rs2	rs1	funct 3	rd	Opcode

Explanation:

→ opcode - Specifies operation type
→ rd. Destination Register
→ rs1, rs2 - Source register
→ funct 3 & funct 7 - Define exact operation
→ fixed length (32-bit) format simplifies decoding.

Ans-3



Explanation

- Data is read from source registers.
- Sent to ALU for operation
- Result is written back to destination register
- Enable fast computation without memory access.